

ADITYA (ADI) RAVI

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EDUCATION

University of Wisconsin – Madison | Madison, WI

Major: Master of Science in Physics – Quantum Computing

2025 – 2027

The Pennsylvania State University | State College, PA

Major: Bachelor of Science in Industrial Engineering

2018 – 2022

SKILLS & RELEVANT COURSEWORK

Quantum Theory
Qubit Tune-up & Programming
Cold atom trapping and sensing
Quantum Algorithms

Data Analysis & Visualization
SQL, KQL, R, Excel
Power BI, Google Analytics, Grafana
A/B Testing, User testing

Technical Skills & Tools
Qiskit, Python, MATLAB,
Azure, REST APIs, Postman
Docker, Kubernetes, ETL

Quantum Mechanics | Quantum Information Science (QIS) | Atomic Physics | Quantum Algorithms | Linear Algebra
Engineering Analytics | Statistics & Probability |

RESEARCH EXPERIENCE

Otten Group – Theoretical Quantum Information Science | Madison, WI

Jan 2026 - Current

Research Assistant | Advisor: Dr. Matthew Otten

Focus: Investigating quantum algorithm design in the early fault-tolerant regime, where devices are partially error-corrected and constrained by limited logical qubits, magic-state throughput, imperfect decoding, and finite error-correction cycle times.

- Studying how algorithm parameters such as precision, Trotter step count, circuit depth, and approximation thresholds should be adjusted when full fault tolerance is unavailable or resource limited.
- Comparing candidate algorithms like QPE, Hamiltonian simulation, and variational or hybrid circuits to identify workloads suited for early fault-tolerant quantum hardware.

Quantum Circuit Evaluator Tool

Built a full-stack open source Qiskit benchmarking platform to evaluate quantum-circuit transpilation across configurable NISQ and fault-tolerant architecture assumptions.

- Developed OpenQASM ingestion and circuit-standardization workflows for consistent evaluation.
- Modeled native gate sets, connectivity topologies, operation errors, durations, and modular communication constraints.
- Evaluating resource-estimation frameworks like BenchQ along with adding a Clifford+T decomposition to study routing overhead effects on non-Clifford cost for QPE.

Sinclair Quantum Lab – Neutral Atom Quantum Computing | Madison, WI**Aug 2025 – April 2026****Research Assistant | Advisor: Dr. Josiah Sinclair**

Contributing as an early member to the lab's experimental infrastructure for modular neutral-atom quantum computing, including atom trapping, laser stabilization, and scalable hardware systems.

- Used Zemax to model beam propagation and optimize the optical layout for a pseudo magneto-optical trap (PMOT), defining alignment and stability requirements for atom trapping and cooling.
- Developed the lab's initial end-to-end data pipeline, enabling continuous streaming of magnetic-field and other sensor readings to a Synology NAS and Influx DB backend with real-time Grafana visualization.
- Designed, modeled, and built a three-axis Helmholtz coil system to actively stabilize the ambient magnetic field against long-term drift. Characterized field uniformity and multi-day drift behavior, then implemented a Python-based PID control loop to close the feedback and maintain field stability.

PROFESSIONAL EXPERIENCE**CARIAD Inc. – A Volkswagen Group Company | Seattle, WA****2022 - 2024****Product Manager***May 2024 – December 2024*

Led product strategy for a new team, steering the development and deployment of cloud-native software updates across VW Group vehicles, impacting 2 million vehicles globally. Defined product increment OKRs with engineering, managed the backlog, prioritized tasks, and engaged stakeholders to optimize outcomes. Tracked key metrics to ensure continuous product improvement and success.

- Facilitated a seamless product transition by driving and optimizing knowledge transfers between US and German stakeholders using the Learn, Shadow, Lead and Operate model, completing the transition ahead of schedule.
- Defined KPIs and metrics across the product portfolio, improving success measurement accuracy by 25%
- Identified and documented 20+ customer facing use cases, supported by the platform.
- Developed a product roadmap to consolidate functionalities with the existing product stack, reducing redundancies and aligning with business objectives.

Product Analyst*Feb 2023 - May 2024*

Drove data-based decision making across the product portfolio by analyzing API and other product usage. Designed and implemented dashboards with Grafana, Power BI, and Google Analytics, enabling data-driven decisions and product insights.

- Streamlined the communication process for handling feature requests from external partners, resulting in a 15% improvement in response time and overall efficiency.
- Analyzed auth-token approval times for external partners using SQL and KQL, identifying bottlenecks that led to a 50% reduction.
- Helped implement a Deployment and Release checklist resulting in 100% adoption and consistency across all agile teams.
- Defined user journeys and user paths for onboarding new customers, designing experiments and implementing them with the UX team.

Customer Success Engineer – Internship*Jun 2022 – Aug 2022*

Managed customer and stakeholder relationship across product development teams, building apps and services on the Volkswagen Automotive Cloud Platform.

- Outlined an end-to-end new user onboarding journey, identifying key challenges and issues in the process.
- Learned and applied agile methodologies to identifying key challenges and shaping use cases to inform the creation of user personas.